Nonsingularity of the periodic partial-sums system with arbitrary support minima

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MathDB 375984. Full solution of Grigutis–Petkelis Conjecture 1.

Task 1: statement and the source conventions

Let independent integer increments have period N in distribution, minimum support points m_j, and

$$D = \sum_{j=1}^{N} m_j < 0, \qquad \mu = \mathbb{E}\sum_{j=1}^{N} X_j < 0, \qquad \mathbb{P}(X_j = m_j) > 0.$$

Indices on phases are cyclic. Write $\mathcal{M}_j = \max(0, X_j, X_j + X_{j+1}, \ldots)$ and $\pi_i^{(j)} = \mathbb{P}(\mathcal{M}_j = i)$. The source constructs a system for its initial probabilities by evaluating generating-function numerators at the nonunit roots of $G_{\sum X_j}(z) = 1$, and appending a normalization equation [1, (2.7), pp. 7–9]. It conjectures nonsingularity.

Two source conventions matter. A root of multiplicity q gives derivative constraints of orders $0, \ldots, q-1$ [1, Note 1]. Probabilities forced to be zero by support bounds are removed, as in the mixed-sign example [1, pp. 17–18]. The printed block dimensions $-m_j$ apply directly when all $m_j \le 0$; for general integer minima the forced-zero elimination below specifies the $-D$ actual unknowns.

Theorem 1. *Under the assumptions above, the source's coefficient system, with these support and multiplicity conventions, is nonsingular. This holds for every period, arbitrary integer support minima, all multiplicities of interior roots, and roots on the unit circle.*

1. Reduce all support minima to nonpositive values

Define deterministic heights

$$\ell_j = \max\left(0, m_j, m_j + m_{j+1}, \ldots, m_j + \cdots + m_{j+N-1}\right).$$

Since $D < 0$, an extra full period cannot increase a partial sum of the minima. Consequently

$$\ell_j = \max(0, m_j + \ell_{j+1}), \qquad \mathcal{M}_j \ge \ell_j.$$

Set

$$k_j = \ell_j - m_j - \ell_{j+1} \ge 0, \quad C = \sum_j k_j = -D > 0, \quad Y_j = X_j + \ell_{j+1} - \ell_j, \quad Z_j = Y_j + k_j = X_j - m_j. \tag{1}$$

Here $Z_j \geq 0$ and $p_{j,0} := \mathbb{P}(Z_j = 0) > 0$. The transformed increments have minima $-k_j$ and the same period sum as the original increments. Their means are finite: each increment is bounded below, and their total mean is finite and negative. In particular $\sum_j \mathbb{E}Z_j = C + \mu < C$.

The translated maxima $L_j = \mathcal{M}_j - \ell_j$ obey

$$L_j \stackrel{d}{=} \max(0, Y_j + L_{j+1}). \tag{2}$$

Indeed the original recursion gives $L_j = \max(-\ell_j, Y_j + L_{j+1})$ in distribution. If $\ell_j = 0$, this is (2). If $\ell_j > 0$, then $k_j = 0$ and $Y_j \geq 0$, so both maxima equal $Y_j + L_{j+1}$.

In phase j, the source initially uses $\pi_i^{(j+1)}$, $0 \leq i < -m_j$. Those with $i < \ell_{j+1}$ are zero. If $\ell_j > 0$, none remain. If $\ell_j = 0$, the remaining indices are precisely

$$i = \ell_{j+1} + u, \quad 0 \leq u < k_j. \tag{3}$$

Thus exactly $\sum_j k_j = C = -D$ unknowns remain in all cases. When every $m_j \leq 0$, all heights vanish and this is the displayed source matrix without any change of variables.

2. An equivalent description of every matrix row

Put $g_j(z) = \mathbb{E}z^{Z_j}$, $F_j(t) = \mathbb{P}(Z_j \leq t)$. For an arbitrary complex vector $v = (v_{j,u})_{0 \leq u < k_j}$, define

$$Q_j(z) = \sum_{u=0}^{k_j-1} v_{j,u} \sum_{t=u}^{k_j-1} z^t F_j(t-u), \qquad Q_j = 0 \text{ if } k_j = 0. \tag{4}$$

The map from v_j to the coefficients of Q_j is triangular with positive diagonal $F_j(0) = p_{j,0}$. Let

$$K(z) = \prod_{j=1}^{N} g_j(z) - z^C, \qquad H_j(z) = \sum_{r=0}^{N-1} z^{C - \sum_{t=0}^{r} k_{j+t}} \left(\prod_{t=0}^{r-1} g_{j+t}(z) \right) Q_{j+r}(z). \tag{5}$$

Empty products equal one. Every exponent in (5) is nonnegative. The root-count result used in [1], namely [2, Lemma 4] applied with one aggregate variable $\sum_j Z_j$ and premium C, gives C zeros of K in the closed unit disk, counted with multiplicity. One is 1. None is zero, since $K(0) = \prod_j p_{j,0} > 0$.

Up to invertible row operations, the homogeneous system is

$$H_N^{(t)}(\alpha) = 0 \quad (K(\alpha) = 0,\ \alpha \neq 1,\ 0 \leq t < \mathrm{mult}_K(\alpha)), \qquad \sum_j Q_j(1) = 0. \tag{6}$$

Here and below boundary roots will be shown to be simple. To verify the exact source correspondence, let B_j^{old} be the Laurent polynomial in [1, Proposition 1]. After substituting (3) into its definition, direct index shifting gives

$$B_j^{\mathrm{old}}(z) = z^{\ell_j - k_j} Q_j(z), \qquad G_{X_j}(z) = z^{\ell_j - \ell_{j+1} - k_j} g_j(z). \tag{7}$$

The first equality is also true when $\ell_j > 0$, since both sides vanish. The numerator in the source's phase-N row, with block order $N, 1, \ldots, N-1$, is therefore

$$B_N^{\mathrm{old}} + G_{X_N} B_1^{\mathrm{old}} + G_{X_N} G_{X_1} B_2^{\mathrm{old}} + \cdots = z^{\ell_N - C} H_N(z). \tag{8}$$

At a nonzero root, multiplication by $z^{\ell_N - C}$ changes a vector of derivatives through order $q-1$ by an invertible triangular matrix. The factor $1 - z$ in the source identity is likewise invertible at $\alpha \neq 1$. At $z = 1$, (7) gives the same normalization row $\sum_j Q_j(1)$, with right-hand side $-\mu$. This proves (6), including the confluent rows.

3. Kernel cancellation gives decaying coefficients

Suppose (6) holds. Telescoping (5) gives

$$g_j H_{j+1} - z^{k_j} H_j = K Q_j, \qquad 1 \le j \le N. \tag{9}$$

At any nonzero root α of K, all $g_j(\alpha)$ are nonzero because their product is α^C. Equation (9) propagates the prescribed vanishing order from H_N to all H_j. Also $H_j(1) = \sum_r Q_r(1) = 0$. It follows that $T_j = H_j/K$ is holomorphic in the open unit disk.

We check continuity on its boundary without imposing exponential tails or higher moments. Finite first moments imply that g_j and g_j' are continuous on the closed disk, by uniform absolute convergence of their series. The same holds for K, H_j and their first derivatives.

If $|\alpha| = 1$ and $K(\alpha) = 0$, let $Z = \sum_j Z_j$. Equality in $|\mathbb{E}\alpha^Z| \le 1$, together with $\mathbb{P}(Z = 0) > 0$, forces $\alpha^r = 1$ on the support of Z, and $\alpha^C = 1$. Therefore

$$K'(\alpha) = \alpha^{-1}(\mathbb{E}Z - C) = \alpha^{-1}\mu \neq 0. \tag{10}$$

All boundary roots are simple. If a function f and f' are continuous on the closed disk and $f(\alpha) = 0$, integration along the segment from α to z shows

$$\frac{f(z)}{z - \alpha} \longrightarrow f'(\alpha) \quad \text{as } z \to \alpha \text{ within the disk.}$$

Apply this to H_j, K; (10) proves continuous extension of T_j at every boundary root.

Consequently the Taylor coefficients $h_j(u) = [z^u]T_j(z)$ tend to zero. For completeness, if T is holomorphic in the disk and continuous on its closure, Cauchy's estimate bounds its coefficients by $M = \max|T|$. For $0 < \rho < 1$, apply that estimate to $T(z) - T(\rho z)$ to get

$$|h(u)| \le M\rho^u + \max_{|z|\le 1} |T(z) - T(\rho z)|.$$

First let $u \to \infty$, then $\rho \uparrow 1$. Uniform continuity proves the assertion.

4. A cyclic maximum principle finishes the proof

Dividing (9) by K gives

$$g_j T_{j+1} - z^{k_j} T_j = Q_j.$$

Since $\deg Q_j < k_j$ (with $Q_j = 0$ when $k_j = 0$), coefficient comparison yields, for every $u \ge 0$,

$$h_j(u) = \sum_{r=0}^{u+k_j} p_{j,r}\, h_{j+1}(u + k_j - r). \tag{11}$$

Here $p_{j,r} = \mathbb{P}(Z_j = r)$. If these finitely many decaying sequences were not all zero, their joint supremum in modulus would be a positive attained maximum M_*, say $|h_j(u)| = M_*$. The right side of (11) has modulus at most M_*, because the weights are nonnegative and sum to at most one. Since $p_{j,0} > 0$, equality forces $|h_{j+1}(u + k_j)| = M_*$. Repeating this implication for a complete period gives $|h_j(u + C)| = M_*$, then $|h_j(u + qC)| = M_*$ for every q. As $C > 0$, this contradicts decay.

Thus all T_j, all Q_j, and, by the triangular invertibility in (4), all $v_{j,u}$ vanish. The square system of C rows has trivial kernel. Equations (7)–(8) transfer this conclusion to the original system, proving Theorem 1.

Scope and attribution

No condition that the source's boundary terms $b_k(x)$ vanish was used. The proof covers unequal negative minima, zero minima, positive minima after the exact support reduction, and every root multiplicity allowed by Note 1. It does not treat duplicated evaluation rows as the intended repeated-root system.

The root count is the established result [2, Lemma 4]. The analytic cancellation and maximum principle extend the earlier campaign proof for MathDB 357004 [3], whose theorem assumed a common positive premium and simple nontrivial roots. That earlier result is credited as the method's source. The additional support-height reduction, variable and zero shifts, and all confluent cases are proved here.

Tasks 2–5: proposed mathematical follow-ups

Task 2: Obtain quantitative lower bounds for the smallest singular value of the normalized confluent matrix in terms of the drift, minimum masses, root separation, and moment bounds.

Task 3: Extend the uniqueness theorem to a finite-state Markov environment and identify the appropriate drift and irreducibility conditions for a matrix-valued kernel.

Task 4: Determine the rank and compatibility conditions at zero period drift, including the degenerate case of an identically zero period sum.

Task 5: Construct a stable algorithm that avoids explicit root locations while computing the same initial probabilities, with rigorous error bounds for unbounded increment distributions.

References

[1] A. Grigutis and J. Petkelis, *The limit law of the maximum of discrete partial-sums distribution II*, arXiv:2607.11275v1 (13 July 2026), Conjecture 1, Note 1, and Example 1.

[2] S. Gervé and A. Grigutis, *Distribution of shifted discrete random walk generated by distinct random variables and applications in ruin theory*, Modern Stochastics: Theory and Applications **11**(3) (2024), 323–357, Lemma 4, pp. 338–339. doi:10.15559/24-VMSTA249.

[3] Principia Math In Progress, *Nonsingularity of the periodic random-walk system*, MathDB 357004, 18 September 2026. Earlier campaign proof.